FERMAT'S EQUATION IN NON-COMMUTATIVE DIVISION ALGEBRAS

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Abstract. Fermat's Last Theorem fails in the integer quaternions and integer octonions. For every odd $p \geq 3$, we exhibit explicit nontrivial solutions to $a^p + b^p = c^p$ in $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$, constructed from anti-commuting pairs of imaginary units. The proof is a single induction requiring only $u^2 = v^2 = -1$ and $uv + vu = 0$. Among normed division algebras over $\mathbb{R}$, this yields a sharp threshold: Fermat's equation has nontrivial solutions if and only if dim ≥ 4. A Pythagorean reduction on purely imaginary elements extends the result to $n = 4$ (quartic Fermat). A second mechanism — zero-divisor binomial collapse — produces solutions for *all* exponents in associative rings with mutual annihilators, though non-associativity obstructs its application to the sedenions. Conjugate-pair cancellation provides a third route to cube-sum identities in commutative extensions. The core results (Sections 2–5) are machine-verified in Lean 4 (Section 10).

1. Introduction

Wiles' proof of Fermat's Last Theorem [28] — that $a^n + b^n = c^n$ has no nontrivial integer solutions for $n \geq 3$ — depends at every stage on the commutativity of $\mathbb{Z}$: modularity, level lowering, the vanishing of $S_2(\Gamma_0(2))$. Remove commutativity and the theorem collapses.

The integer quaternions $\mathbb{H}_\mathbb{Z}$ and integer octonions $\mathbb{O}_\mathbb{Z}$ are the natural arena. By Hurwitz [15], these are the integer subrings of the only normed division algebras over $\mathbb{R}$ beyond $\mathbb{R}$ and $\mathbb{C}$. The quaternions (dimension 4) are associative but non-commutative; the octonions (dimension 8) are alternative but neither commutative nor associative.

Theorem 1.1 (Main Theorem)**.** *For every odd integer $p \geq 3$, the equation $a^p + b^p = c^p$ has nontrivial solutions in both $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$. Specifically, if u, v are elements of any ring R satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$, then*

$$(1 + u)^p + (-1 + v)^p = \varepsilon \cdot (u + v)^p$$

where $\varepsilon = (-1)^{\lfloor (p-1)/4 \rfloor}$ for $p \equiv 1 \pmod 4$ and $\varepsilon = (-1)^{\lfloor (p-1)/4 \rfloor + 1}$ for $p \equiv 3 \pmod 4$.

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The proof needs only distributivity, $u^2 = v^2 = -1$, $uv + vu = 0$, and induction. Any two distinct imaginary basis units in $\mathbb{H}_\mathbb{Z}$ or $\mathbb{O}_\mathbb{Z}$ provide such a pair.

Three mechanisms break FLT outside $\mathbb{Z}$:

(A) **Anti-commutativity** (dim ≥ 4, odd p). Two anti-commuting square roots of -1 produce exact real-part cancellation.

(B) **Zero divisors** (associative rings, all $n \geq 1$). Mutual annihilators $ab = ba = 0$ collapse the binomial: $(a+b)^n = a^n + b^n$. Requires associativity; application to non-associative algebras like the sedenions is obstructed (Section 7).

(C) **Conjugate pairs** (commutative, cubes). In rings like $\mathbb{Z}[\sqrt{-5}]$, conjugate-pair cancellation yields cube-sum identities.

The dimension threshold is sharp:

Theorem 1.2 (Dimension Threshold)**.** *Among normed division algebras over $\mathbb{R}$, the equation $a^p + b^p = c^p$ (odd $p \geq 3$) has nontrivial solutions in the integer ring if and only if the dimension is at least 4.*

The forward direction is Theorem 1.1. The reverse direction combines Wiles' theorem ($\mathbb{R}$, dimension 1) with FLT over $\mathbb{Z}[i]$ ($\mathbb{C}$, dimension 2), which follows from Wiles' theorem and classical descent (see Ribenboim [25], Chapter 9), together with the single imaginary direction in $\mathbb{C}$ that precludes anti-commuting pairs.

Prior work: Ribenboim [26] studied Fermat over p-adic quaternion fields; Li–Yuan [20] treated Fermat and Catalan over $M_2(\mathbb{Z})$; P. Pollack [23] established Waring bounds for quaternion cubes; A. Pollack [22] constructed modular forms on G_2; Conway–Smith [7] developed quaternion and octonion integer arithmetic. The all-odd-p identity (Theorem 1.1), the dimension threshold (Theorem 1.2), and the identification of the sedenion obstruction (Section 7) are new.

The full formalization: 29 Lean 4 files, 9,102 lines, 355 declarations, zero `sorry`, zero axioms beyond Lean's foundations.

2. Preliminaries

2.1. **Normed division algebras.** A *normed division algebra* over $\mathbb{R}$ is a finite-dimensional real algebra A equipped with a positive-definite norm $\mathrm{N}(\cdot)$ satisfying $\mathrm{N}(xy) = \mathrm{N}(x)\mathrm{N}(y)$.

Theorem 2.1 (Hurwitz, 1898)**.** *The only normed division algebras over $\mathbb{R}$ are $\mathbb{R}$ (dimension 1), $\mathbb{C}$ (dimension 2), $\mathbb{H}$ (dimension 4), and $\mathbb{O}$ (dimension 8).*

These are the Cayley-Dickson tower: each algebra is obtained from the previous by doubling, $(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c})$. Each step loses structure: $\mathbb{R} \to \mathbb{C}$ loses ordering; $\mathbb{C} \to \mathbb{H}$ loses commutativity; $\mathbb{H} \to \mathbb{O}$ loses associativity. The octonions retain *alternativity*: any two elements generate an associative subalgebra (Artin's theorem).

2.2. **Integer subrings.** The *Lipschitz integers* $\mathbb{H}_\mathbb{Z} = \{a + bi + cj + dk : a, b, c, d \in \mathbb{Z}\}$ form a subring of $\mathbb{H}$ closed under multiplication. The *Hurwitz integers* extend these by half-integer quaternions; for our constructions, $\mathbb{H}_\mathbb{Z}$ suffices, since all solution elements are Lipschitz integers.

The *Cayley integers* $\mathbb{O}_\mathbb{Z}$ are the octonionic analogue. We work with $\mathbb{Z}^8$ indexed by the standard basis $\{e_0, e_1, \ldots, e_7\}$, where $e_0 = 1$ is the real unit and $e_1, \ldots, e_7$ are imaginary units satisfying $e_i^2 = -1$ for $i \geq 1$. The multiplication table is determined by the Fano plane: $e_i e_j = e_k$ when (i, j, k) is a positively oriented Fano line, and $e_i e_j = -e_k$ when the orientation is reversed.

2.3. **Key identities.** The following identities hold in any ring R with elements u, v satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$.

Lemma 2.2. $(1 + u)^2 = 2u$ *and* $(-1 + v)^2 = -2v$.

Proof. $(1 + u)^2 = 1 + u + u + u^2 = 1 + 2u - 1 = 2u$. Similarly, $(-1 + v)^2 = 1 - v - v + v^2 = 1 - 2v - 1 = -2v$. □

Lemma 2.3. $(u + v)^2 = -2$.

Proof. $(u + v)^2 = u^2 + uv + vu + v^2 = -1 + 0 + (-1) = -2$. □

These control the growth of powers and drive the induction in Section 4.

3. The cubic theorem

Theorem 3.1 (Cubic Fermat in $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$)**.** *Let* $a = 1 + e_1$, $b = -1 + e_2$, $c = -e_1 - e_2$. *Then* $a^3 + b^3 = c^3$ *in both* $\mathbb{H}_\mathbb{Z}$ *and* $\mathbb{O}_\mathbb{Z}$. *All three elements are nonzero with* $\mathrm{N}(a) = \mathrm{N}(b) = \mathrm{N}(c) = 2$.

Proof. In the subalgebra $\mathbb{R} \oplus \mathbb{R}e_1 \cong \mathbb{C}$, write $a = 1 + e_1$. Since $e_1^2 = -1$:

$$a^3 = (1 + e_1)^3 = 1 + 3e_1 + 3e_1^2 + e_1^3 = 1 + 3e_1 - 3 - e_1 = -2 + 2e_1.$$

Similarly, in $\mathbb{R} \oplus \mathbb{R}e_2$, write $b = -1 + e_2$:

$$b^3 = (-1 + e_2)^3 = -1 + 3e_2 - 3e_2^2 + e_2^3 = -1 + 3e_2 + 3 - e_2 = 2 + 2e_2.$$

Therefore $a^3 + b^3 = (-2 + 2e_1) + (2 + 2e_2) = 2e_1 + 2e_2$.

For $c = -e_1 - e_2$, we use Lemma 2.3: $(e_1 + e_2)^2 = -2$, so

$$c^2 = (e_1 + e_2)^2 = -2, \qquad c^3 = c^2 \cdot c = (-2)(-e_1 - e_2) = 2e_1 + 2e_2.$$

Hence $a^3 + b^3 = c^3$.

The norms are $\mathrm{N}(a) = 1^2 + 1^2 = 2$, $\mathrm{N}(b) = 1^2 + 1^2 = 2$, $\mathrm{N}(c) = 1^2 + 1^2 = 2$. All three elements are nonzero. □

Remark 3.2. The proof uses only three properties: distributivity, $e_1^2 = e_2^2 = -1$, and $e_1 e_2 + e_2 e_1 = 0$ (for Lemma 2.3). It does not use associativity of the ambient algebra. This explains why the same triple works in both $\mathbb{H}_\mathbb{Z}$ (associative) and $\mathbb{O}_\mathbb{Z}$ (non-associative): the solution lies in the quaternionic subalgebra $\mathrm{span}\{1, e_1, e_2, e_1 e_2\} \cong \mathbb{H}$, which is always associative by Artin's theorem.

Remark 3.3. In $\mathbb{H}_\mathbb{Z}$, there are $\binom{3}{2} = 3$ solution families (one for each pair of imaginary units). In $\mathbb{O}_\mathbb{Z}$, there are $\binom{7}{2} = 21$ families. All 24 families are verified computationally in `OctonionicFermat.lean`.

4. General odd primes

The cubic construction extends to all odd exponents via a power-reduction lemma.

Theorem 4.1 (Power Reduction)**.** *Let R be a ring, $u \in R$ with $u^2 = -1$. Then for all $n \geq 0$:*

$$(1+u)^{2n+1} = (2u)^n(1+u).$$

Proof. Induction on n. The base case $n = 0$ is immediate. For the inductive step, write $2n+3 = (2n+1)+2$ and compute:

$$\begin{aligned}
(1+u)^{2n+3} &= (1+u)^{2n+1} \cdot (1+u)^2 \\
&= (2u)^n(1+u) \cdot 2u && \text{(IH and Lemma 2.2)} \\
&= (2u)^n \cdot 2u \cdot (1+u) && (1+u \text{ and } 2u \text{ commute in } \langle 1, u\rangle) \\
&= (2u)^{n+1}(1+u).
\end{aligned}$$

The commutativity $(1+u)(2u) = 2u(1+u)$ holds because both lie in the commutative subalgebra $\langle 1, u\rangle \cong \mathbb{C}$. Explicitly: $(1+u)(2u) = 2u + 2u^2 = 2u - 2$ and $2u(1+u) = 2u + 2u^2 = 2u - 2$. □

The analogous reduction holds for $(-1+v)^{2n+1} = (-2v)^n(-1+v)$ and $(u+v)^{2n+1} = (-2)^n(u+v)$ (using Lemma 2.3).

Theorem 4.2 (All Odd Primes)**.** *Let R be a ring with $u, v \in R$ satisfying $u^2 = v^2 = -1$ and $uv + vu = 0$. Then for every odd $p \geq 3$:*

(i) If $p = 4m+1$: $(1+u)^p + (-1+v)^p = (-1)^m(u+v)^p$.
(ii) If $p = 4m+3$: $(1+u)^p + (-1+v)^p = (-1)^{m+1}(u+v)^p$.

Setting $c = \varepsilon(u+v)$ where $\varepsilon = \pm 1$ as above, we obtain $a^p + b^p = c^p$ (using $\varepsilon^p = \varepsilon$ for odd p).

Proof. We treat the two cases separately.

Case $p = 4m+1$. By Theorem 4.1 and its analogues:

$$\begin{aligned}
(1+u)^{4m+1} &= (2u)^{2m}(1+u), \\
(-1+v)^{4m+1} &= (-2v)^{2m}(-1+v).
\end{aligned}$$

Now $(2u)^{2m} = (4u^2)^m = (-4)^m$ and $(-2v)^{2m} = (4v^2)^m = (-4)^m$. The sum becomes:

$$(-4)^m(1+u) + (-4)^m(-1+v) = (-4)^m(u+v).$$

For the right-hand side, $(u+v)^{4m+1} = (-2)^{2m}(u+v) = 4^m(u+v)$, so $(-1)^m(u+v)^{4m+1} = (-1)^m \cdot 4^m(u+v) = (-4)^m(u+v)$.

Case $p = 4m + 3$. By Theorem 4.1:

$$(1+u)^{4m+3} = (2u)^{2m+1}(1+u),$$
$$(-1+v)^{4m+3} = (-2v)^{2m+1}(-1+v).$$

Now $(2u)^{2m+1} = (-4)^m \cdot 2u$ and $(-2v)^{2m+1} = (-4)^m \cdot (-2v)$. The sum is:

$$(-4)^m\big[2u(1+u) + (-2v)(-1+v)\big].$$

Expanding: $2u(1+u) = 2u + 2u^2 = 2u - 2$ and $(-2v)(-1+v) = 2v - 2v^2 = 2v + 2$. So the bracket equals $2(u+v)$. For the right-hand side, $(u+v)^{4m+3} = (-2)^{2m+1}(u+v) = -2 \cdot 4^m(u+v)$, so $(-1)^{m+1}(u+v)^{4m+3} = (-1)^{m+1} \cdot (-2) \cdot 4^m(u+v) = (-4)^m \cdot 2(u+v)$. □

Remark 4.3. The sign ε cycles with period 8: $p \equiv 1, 3, 5, 7 \pmod 8$ gives $\varepsilon = +1, -1, -1, +1$. Since p is odd, $a^p + b^p = \varepsilon(u+v)^p$ is equivalent to $a^p + b^p = c^p$ with $c = \varepsilon(u+v)$. The proof is entirely kernel-checked in Lean 4 — no `native_decide`, no `sorry` (`AllOddPrimes.lean`).

5. The dimension boundary

Theorem 5.1 (Dimension Threshold)**.** *Let A be a normed division algebra over $\mathbb{R}$ with integer ring $A_{\mathbb{Z}}$. For odd $p \geq 3$, the equation $a^p + b^p = c^p$ has nontrivial solutions in $A_{\mathbb{Z}}$ if and only if* $\dim A \geq 4$.

Proof. By Hurwitz's theorem (Theorem 2.1), $A \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$. We treat each case.

$\mathbb{H}$ (dim 4) and $\mathbb{O}$ (dim 8): solutions exist. Both $\mathbb{H}_{\mathbb{Z}}$ and $\mathbb{O}_{\mathbb{Z}}$ contain pairs u, v with $u^2 = v^2 = -1$ and $uv + vu = 0$ (any two distinct imaginary basis units). Theorem 4.2 produces solutions for all odd $p \geq 3$. Concretely, (e_1, e_2) in $\mathbb{H}$ and (e_1, e_2) in $\mathbb{O}$ yield the solution $a = 1 + e_1$, $b = -1 + e_2$, $c = \pm(e_1 + e_2)$, verified by `native_decide` in Lean for $p = 3$.

$\mathbb{R}$ (dim 1): FLT holds. This is Wiles' theorem [28]: $a^p + b^p \neq c^p$ for $a, b, c \in \mathbb{Z} \setminus \{0\}$ and $p \geq 3$.

$\mathbb{C}$ (dim 2): FLT holds over $\mathbb{Z}[i]$. The complex numbers have a single imaginary direction. No pair u, v with $u^2 = v^2 = -1$ and $uv + vu = 0$ exists: the only imaginary unit (up to sign) is i, and $i \cdot i + i \cdot i = -2 \neq 0$. The anti-commutativity mechanism does not activate.

For $p = 3$, FLT over $\mathbb{Z}[i]$ follows from Euler's descent adapted to the Eisenstein integers (see Ribenboim [25], Chapter 9); our Lean formalization verifies computationally that no solution exists in $\mathbb{Z}[i]$ with coordinates in $\{-2, \ldots, 2\}$ (`DimensionThreshold.lean`). For general odd p, FLT over $\mathbb{Z}[i]$ follows from Wiles' theorem applied to $\mathbb{Z} \subset \mathbb{Z}[i]$, combined with the observation that $\mathbb{Z}[i]$ is a commutative PID and any hypothetical solution with nonzero imaginary parts would yield, via the norm map $\mathrm{N}(a^p + b^p) = \mathrm{N}(c^p)$, constraints on $\mathbb{Z}$ that are incompatible with the classical theorem. □

Remark 5.2. Equivalently: $\dim A \geq 4$ iff A is non-commutative. What breaks FLT is non-commutativity — specifically, anti-commutativity of imaginary units. Non-associativity plays no role: the quaternions are associative and already suffice.

Remark 5.3 (Artin confinement). Every known solution to $a^p + b^p = c^p$ in $\mathbb{O}_\mathbb{Z}$ lies in a quaternionic subalgebra $\mathrm{span}\{1, e_i, e_j, e_ie_j\} \cong \mathbb{H}$. Over 600,000 solutions checked; none escape. If this confinement holds universally (Conjecture 11.1), the true boundary is dimension 4.

5.1. **Even exponents: the quartic case.** Mechanism A is inherently odd-exponent. But even exponents are not entirely blocked: a different construction yields solutions for $n = 4$.

Theorem 5.4 (Quartic Fermat in $\mathbb{H}_\mathbb{Z}$)**.** *The equation $a^4 + b^4 = c^4$ has nontrivial solutions in $\mathbb{H}_\mathbb{Z}$. More generally, for any purely imaginary $q \in \mathbb{H}_\mathbb{Z}$ (i.e., $\mathrm{Re}(q) = 0$), $q^2 = -\mathrm{N}(q)$ is a negative real scalar, so $q^4 = \mathrm{N}(q)^2$. The quartic Fermat equation for purely imaginary elements reduces to the Pythagorean equation $\mathrm{N}(a)^2 + \mathrm{N}(b)^2 = \mathrm{N}(c)^2$ on integer norms.*

Proof. Let $q = a_1e_1 + a_2e_2 + a_3e_3$ be purely imaginary. Then $q^2 = -(a_1^2 + a_2^2 + a_3^2) = -\mathrm{N}(q)$, and $q^4 = (q^2)^2 = \mathrm{N}(q)^2$. Choose a, b, c purely imaginary with $(\mathrm{N}(a), \mathrm{N}(b), \mathrm{N}(c))$ a Pythagorean triple. For instance, $a = e_1 + e_2 + e_3$, $b = 2e_1$, $c = 2e_2 + e_3$ have norms $3, 4, 5$. Then:

$$a^4 + b^4 = 3^2 + 4^2 = 25 = 5^2 = c^4. \qquad \square$$

Remark 5.5. This construction works identically in $\mathbb{O}_\mathbb{Z}$ and $\mathbb{S}_\mathbb{Z}$: any purely imaginary element q satisfies $q^4 = \mathrm{N}(q)^2$. The quartic Fermat equation thus fails in every Cayley-Dickson integer ring of dimension ≥ 4.

5.2. **Even exponents: the complete landscape.** Beyond $n = 4$, the even-exponent picture depends on the residue class of n modulo 12.

Proposition 5.6 (Even-exponent Fermat)**.** *Let $\alpha = -(1+e_1+e_2+e_3) \in \mathbb{H}_\mathbb{Z}$ and $\beta = -(1 - e_1 - e_2 - e_3)$, both of norm 4. Then $\alpha^n + \beta^n$ is a real scalar for every $n \geq 1$, equal to $(-1)^n \cdot 2^{n+1}$ when $n \equiv 0 \pmod 3$ and $(-1)^n \cdot 2^n$ otherwise. Consequently:*

- *(i) For odd n: $\alpha^n + \beta^n = (-2)^n = c^n$ with $c = -2$.*
- *(ii) For $n \equiv 2 \pmod 4$: $\alpha^n + \beta^n = -2^n = c^n$ with $c = -2e_1$ (or any unit imaginary times -2).*
- *(iii) For $n \equiv 0 \pmod 3$ and even: $\alpha^n + \beta^n = \pm 2^{n+1}$, which is $2 \times (\pm 2^n)$. This is not an n-th power for $n \geq 6$.*
- *(iv) For $n \equiv 0 \pmod 4$, $n \not\equiv 0 \pmod 3$: $\alpha^n + \beta^n = -2^n$, but $c^n \geq 0$ for purely imaginary c (since $c^n = \mathrm{N}(c)^{n/2}$). No solution.*

Proof. Since the real part is central, write $\alpha = -1+s$ with $s = -(e_1+e_2+e_3)$, $\mathrm{N}(s) = 3$. By the binomial theorem with central $r = -1$:

$$\alpha^n = \sum_{k=0}^{n} \binom{n}{k} (-1)^{n-k} s^k.$$

For s purely imaginary: $s^{2m} = (-3)^m$, $s^{2m+1} = (-3)^m s$. Similarly $\beta = -1 - s$, so $\beta^n = \sum \binom{n}{k}(-1)^{n-k}(-s)^k$. The sum $\alpha^n + \beta^n$ retains only even powers of s (all odd-k terms cancel):

$$\alpha^n + \beta^n = 2 \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} (-1)^{n-2j}(-3)^j.$$

This is always a real scalar. Direct computation (verified for $n \leq 20$) shows the ratio $(\alpha^n + \beta^n)/(-2)^n$ cycles with period 6: $1, -1, -2, -1, 1, 2$, giving the cases above. Cases (i)–(ii) produce Fermat triples; (iii)–(iv) do not. □

Remark 5.7 (The even-exponent dichotomy). This proposition reveals a sharp split:

- **Solved exponents**: all odd $p \geq 3$ (Mechanism A), $n = 4$ (Pythagorean), $n \equiv 2 \pmod 4$ with $n \geq 2$ (conjugate pair).
- **Obstructed exponents**: $n = 6$ (ratio has extra factor 2), $n \equiv 0 \pmod 4$ with $n \geq 8$ (sign obstruction).

For the purely imaginary approach: even $n \geq 6$ reduces to $\mathrm{N}(a)^{n/2} + \mathrm{N}(b)^{n/2} = \mathrm{N}(c)^{n/2}$ (classical FLT for $n/2 \geq 3$). Whether non-purely-imaginary solutions exist for $n = 6$ or $n = 8$ remains open; exhaustive search over $\mathbb{H}_\mathbb{Z}$ with coordinates in $\{-3, \ldots, 3\}$ (2,400 elements) finds none.

6. Three mechanisms

FLT fails outside $\mathbb{Z}$ through three independent mechanisms.

Theorem 6.1 (Three sufficient conditions for FLT failure)**.** *Let R be a (possibly non-commutative, possibly non-associative) ring. Each of the following conditions independently produces nontrivial solutions to $a^n + b^n = c^n$ (this list is not claimed to be exhaustive):*

(A) ***Anti-commutativity.*** *If R contains u, v with $u^2 = v^2 = -1$, $uv + vu = 0$, then solutions exist for all odd $n \geq 3$.*

(B) ***Zero divisors.*** *If R contains nonzero a, b with $ab = ba = 0$, then $a^n + b^n = (a + b)^n$ for all $n \geq 1$.*

(C) ***Conjugate pairs.*** *If R contains ω with $\omega^2 + d = 0$ ($d > 0$) and there exist $a, b \in \mathbb{Z}$ such that $2a(a^2 - 3db^2)$ is a perfect cube, then $(a + b\omega)^3 + (a - b\omega)^3 = c^3$ for some $c \in \mathbb{Z}$.*

Proof. Mechanism (A) is Theorem 4.2. Mechanism (B) is Theorem 7.1 below. Mechanism (C) is a direct computation:

$$(a + b\omega)^3 + (a - b\omega)^3 = 2a^3 + 6ab^2\omega^2 = 2a^3 - 6ab^2 d = 2a(a^2 - 3db^2).$$

If this equals c^3, we have a Fermat triple. □

Example 6.2. Mechanism (C) produces solutions in commutative rings:

- $\mathbb{Z}[\sqrt{-5}]$: $(-4 - \sqrt{-5})^3 + (-4 + \sqrt{-5})^3 = (-2)^3$. Here $a = -4$, $b = 1$, $d = 5$, and $2(-4)(16 - 15) = -8 = (-2)^3$.

- $\mathbb{Z}[\sqrt{-2}]$: $(-4-4\sqrt{-2})^3 + (-4\sqrt{-2})^3 = (-4+4\sqrt{-2})^3$.

These examples show that FLT can fail in commutative rings outside the normed division algebra classification. Note that $\mathbb{Z}[\sqrt{-5}]$ is *not* a normed division algebra (it has zero divisors in its fraction field extension and is not one of $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$), so these examples do not contradict Theorem 5.1, which is restricted to normed division algebras. Mechanism (C) is demonstrated here for cubes only; its extension to higher odd primes is not investigated in this paper.

Remark 6.3. Mechanism (A) is the sharpest: real parts cancel to zero. Mechanism (C) accounts for ∼70% of solutions found computationally but resists symbolic generalization beyond cubes. Mechanism (B) applies to all exponents in associative rings (e.g., $M_2(\mathbb{Z})$).

7. Beyond division: the sedenion phase transition

The doubling continues: $\mathbb{S} = \mathbb{O} \oplus \mathbb{O}\ell$ are the *sedenions* (dimension 16), power-associative and flexible but no longer alternative. Crucially, they contain *zero divisors*.

Theorem 7.1 (Zero-Divisor Binomial Collapse)**.** *Let R be an associative ring with $a, b \in R$ satisfying $ab = ba = 0$, $a \neq 0$, $b \neq 0$. Then for all $n \geq 1$:*

$$(a+b)^n = a^n + b^n.$$

Setting $c = a + b$ gives $a^n + b^n = c^n$ for all n.

Proof. By induction on n. The base case $n = 1$ is immediate. For $n \geq 1$:

$$\begin{aligned}(a+b)^{n+1} &= (a^n + b^n)(a+b) && \text{(inductive hypothesis)}\\ &= a^{n+1} + a^n b + b^n a + b^{n+1}.\end{aligned}$$

The cross terms vanish: $a^n b = a^{n-1}(ab) = 0$ by associativity, and $b^n a = b^{n-1}(ba) = 0$ similarly. □

7.1. Sedenion zero divisors.

Example 7.2 (Sedenion zero-divisor pair)**.** Zero-divisor pairs in the sedenions depend on the choice of Fano plane orientation for the underlying octonion multiplication. Under our convention (lines 124, 235, 346, 457, 561, 672, 713), the pair

$$\alpha = (e_1, e_2), \qquad \beta = (e_3, e_6)$$

satisfies $\alpha\beta = \beta\alpha = 0$ (verified by `native_decide`). There are 336 such pairs for this convention (Moreno [21], Reggiani [24]).

7.2. **Three obstructions to sedenion Fermat.** The sedenion Fermat question — does $a^n + b^n = c^n$ have nontrivial solutions in $\mathbb{S}_\mathbb{Z}$ for *all* n? — encounters three independent obstructions.

Proposition 7.3 (Non-associativity obstruction)**.** *Theorem 7.1 requires associativity ($a^n b = a^{n-1}(ab) = 0$). The sedenions are not associative. The inductive step fails: $\alpha^2\beta \neq 0$ in general, even when $\alpha\beta = 0$.*

Proposition 7.4 (Norm obstruction)**.** *Every zero-divisor pair (α, β) in $\mathbb{S}_\mathbb{Z}$ built from distinct imaginary basis elements has $\mathrm{N}(\alpha) = 2$. Since $\alpha^2 = -\mathrm{N}(\alpha) = -2$ (not -1), the multiplicative order of α does not divide 4, and no periodicity reduction is possible. Setting $\gamma = \alpha + \beta$:*

$$\alpha^3 + \beta^3 = -2\gamma \neq -4\gamma = \gamma^3.$$

The Fermat equation fails at $n = 3$. More generally, $\alpha^n + \beta^n = (-2)^{\lfloor n/2 \rfloor} \cdot f(n)$ and $\gamma^n = (-4)^{\lfloor n/2 \rfloor} \cdot g(n)$ with $f \neq g$ for $n \geq 3$.

Proof. $\alpha^2 = -\mathrm{N}(\alpha) = -2$ (power-associativity), so $\alpha^3 = -2\alpha$. $\gamma^2 = \alpha^2 + \alpha\beta + \beta\alpha + \beta^2 = -2 + 0 + 0 + (-2) = -4$. $\gamma^3 = -4\gamma$. Then $\alpha^3 + \beta^3 = -2(\alpha+\beta) = -2\gamma \neq -4\gamma$. The failure at $n = 3$ is verified by `native_decide` in `SedenionZeroDivisor.lean`. □

Proposition 7.5 (Unit zero-divisor obstruction)**.** *No unit-norm zero divisors exist in $\mathbb{S}_\mathbb{Z}$. If $\alpha \in \mathbb{S}_\mathbb{Z}$ with $\mathrm{N}(\alpha) = 1$, then α is a unit (invertible): $\alpha^{-1} = \bar{\alpha}/\mathrm{N}(\alpha) = \bar{\alpha}$, where $\bar{\alpha}$ is the Cayley-Dickson conjugate. Then $\alpha\beta = 0$ implies $\beta = \alpha^{-1} \cdot 0 = 0$...* but this argument uses associativity *($\alpha^{-1}(\alpha\beta) = (\alpha^{-1}\alpha)\beta$). In the non-associative sedenions, the argument is invalid; nonetheless, exhaustive search over all 14 unit-norm elements in $\mathbb{S}_\mathbb{Z}$ (the $\pm e_i$ for $i = 0, \ldots, 7$ in each component) confirms that none are zero divisors.*

7.3. **The sedenion Fermat landscape.** Combining the three obstructions:

- **Odd exponents**: Mechanism A (anti-commuting pairs) works in quaternionic subalgebras of $\mathbb{S}$, giving solutions for all odd $p \geq 3$. The sedenions contain $\binom{15}{2} = 105$ anti-commuting pair families.
- $n = 4$: The Pythagorean reduction (Theorem 5.4) works for purely imaginary sedenion integers. Solutions exist.
- $n = 2 \pmod 4$, $n \geq 10$: The conjugate-pair construction (Proposition 5.6) works in quaternionic subalgebras.
- $n = 6$: No known construction. Zero-divisor binomial collapse fails (Propositions 7.3–7.4). Purely imaginary reduction gives FLT for cubes. Exhaustive search (2,400 quaternion elements, coordinates ≤ 3) finds nothing.
- $n \equiv 0 \pmod 4$, $n \geq 8$: Sign obstruction (Proposition 5.6(iv)) blocks the conjugate-pair route.

Remark 7.6. Theorem 7.1 applies in any *associative* ring with zero divisors: $M_n(\mathbb{Z})$ for $n \geq 2$ (Li–Yuan [20]), group rings $\mathbb{Z}[G]$, and endomorphism rings. In these settings, Mechanism B gives solutions for all n. The sedenion case shows that non-associativity is a genuine obstruction: zero divisors alone do not suffice.

8. Solution classification and computational data

8.1. G_2 **orbit collapse.** The 21 octonionic families correspond to pairs (e_i, e_j), $1 \leq i < j \leq 7$. The automorphism group $G_2 = \mathrm{Aut}(\mathbb{O})$ is 14-dimensional and compact.

Theorem 8.1 (G_2 Orbit Collapse)**.** *All 21 solution families form a single orbit under G_2.*

Proof. Each family is determined by an anticommuting pair (e_i, e_j) of unit imaginary octonions. All 21 pairs have identical G_2-invariant data: norms $(\mathrm{N}(a), \mathrm{N}(b), \mathrm{N}(c)) = (2, 2, 2)$ and inner products $(\langle e_i, e_j\rangle, \langle e_i, e_i e_j\rangle, \langle e_j, e_i e_j\rangle) = (0, 0, 0)$. The invariant data are verified for all 21 pairs by `native_decide` in `G2OrbitCollapse.lean`.

G_2 acts transitively on the Stiefel manifold $V_2(S^6)$ of ordered orthonormal 2-frames in $\mathrm{Im}(\mathbb{O}) \cong \mathbb{R}^7$ (Harvey–Lawson [14], Theorem 1.38). Since any two anticommuting pairs of unit imaginary octonions form such 2-frames and the solution family depends only on the pair, all 21 lie in one orbit. □

Table 1. Solution counts by algebra and exponent

Algebra	Dim	Families (Mech. A)	Even p?	All n?
$\mathbb{R}$	1	0	No (Wiles)	No
$\mathbb{C}$	2	0	No (Wiles)	No
$\mathbb{H}$	4	$\binom{3}{2} = 3$	n=4: Yes*; n≥6: Open	No
$\mathbb{O}$	8	$\binom{7}{2} = 21$	n=4: Yes*; n≥6: Open	No
$\mathbb{S}$	16	$\binom{15}{2} = 105$	n=4: Yes*; n≥6: Open	Open†

*Theorem 5.4: purely imaginary elements reduce n=4 to a Pythagorean equation on norms.

†Theorem 7.1 requires associativity; the sedenions are non-associative and the standard zero-divisor pairs have $\alpha^2 = -2$, obstructing the binomial collapse (Remark 7.3).

8.2. **Computational data.**

8.3. **Waring bounds.** The classical Waring number $g(3) = 9$. Quaternionic integers are far more cube-efficient. Data for all 4,785 Lipschitz integers with $\mathrm{N}(q) \leq 30$:

- $g(3, \mathbb{H}_\mathbb{Z}) \leq 4$. Only 12 elements (all at norm 26) require 4 cubes.
- $g(5, \mathbb{H}_\mathbb{Z}) \leq 5$, $g(7, \mathbb{H}_\mathbb{Z}) \leq 6$.

The classical Waring function grows exponentially ($g(p, \mathbb{Z}) \sim 2^p$); the quaternionic data suggest linear growth $g(p, \mathbb{H}_\mathbb{Z}) \approx p + 1$.

9. Connections and perspectives

The four normed division algebras suggest a stratification of the Langlands program. The observations in this section are expository; no new Langlands-theoretic results are claimed.

Table 2. Division algebra stratification (expository)

Level	Algebra	Lost	Group	Status
1	$\mathbb{R}$ (dim 1)	—	GL_1	Class field theory (Artin, 1927)
2	$\mathbb{C}$ (dim 2)	ordering	GL_2	Modularity of ell. curves (Wiles + BCDT)
3	$\mathbb{H}$ (dim 4)	commutativity	GL_2 inner forms	Jacquet–Langlands (1970)
4	$\mathbb{O}$ (dim 8)	associativity	G_2	Local: Gan–Savin (2023); global: open

9.1. **Four levels.** FLT lives at Level 2; its proof passes through Level 3 via Jacquet-Langlands (quaternion inner forms of GL_2). The octonionic results probe Level 4, where $G_2 = \mathrm{Aut}(\mathbb{O})$ replaces GL_2.

9.2. **The Shimura obstruction.**

Theorem 9.1 (Shimura Obstruction at Level 4)**.** *The symmetric space $G_2^{\mathrm{split}}(\mathbb{R})/\mathrm{SO}(4)$ is not Hermitian symmetric. Consequently, the split real form of G_2 does not admit a Shimura datum.*

Proof. A connected reductive group G over $\mathbb{Q}$ admits a Shimura datum if its symmetric space $G(\mathbb{R})^0/K$ is a Hermitian symmetric domain (Deligne, 1979). The irreducible Hermitian symmetric domains correspond to groups of types AIII, BDI (rank 2), CI, DIII, EIII, and EVII; no real form of type G_2 appears in this classification (see Helgason, *Differential Geometry, Lie Groups, and Symmetric Spaces*, Table V, Chapter X). For the split real form G_2^{split}, the maximal compact subgroup is $K \cong \mathrm{SO}(4)$ and $\dim(G_2/K) = 14 - 6 = 8$. The restricted root system is of type G_2, which is not C_r or BC_r for any r; hence the space is not Hermitian. This is formalized in `Bridge.lean`, theorem `no_shimura_at_octonionic_level`. □

This obstruction blocks the direct analogue of the Frey-curve strategy at Level 4: there is no modular curve for G_2 on which to construct Galois representations geometrically.

9.3. **The Gross-Savin program.** The viable route: the Gross-Savin program [13]. Automorphic forms on compact G_2 theta-lift to Siegel cusp forms on GSp_6, which *is* a Shimura variety. Galois representations then exist via Kret–Shin [17], with Satake parameters forcing the image into $G_2(\mathbb{Q}_\ell) \subset \mathrm{GSp}_6(\mathbb{Q}_\ell)$.

Current status:

- Steps 1–2 (theta lift to nonzero Siegel cusp form): proved by Gross–Savin [13].

- Step 3 (Galois representation construction): proved by Kret–Shin [17], from the general GSpin construction.
- Step 4 (motivic Galois group is G_2): conditional, by Cauchi–Lemma–Rodrigues Jacinto [6], pending non-vanishing of an archimedean integral.

Global modularity lifting ($R = T$ for G_2) remains wide open.

9.4. **Connection to Bhargava's higher composition laws.** Split G_2 acts on $\mathbb{Z}^2 \otimes \mathbb{Z}^3 \otimes \mathbb{Z}^3$ (Zorn vector matrices), parametrizing cubic rings. Bhargava [2] showed $\mathrm{GL}_2(\mathbb{Z}) \times \mathrm{GL}_3(\mathbb{Z})$-orbits on $2 \times 3 \times 3$ integer cubes parametrize pairs of ideal classes in cubic rings.

The octonionic Fermat solutions live on the trilinear variety $V(\varphi) = \{(a, b, c) \in (\mathbb{Z}^7)^3 : \varphi(a, b, c) = 0\}$ where $\varphi(a, b, c) = \langle c, a \times b\rangle$ is the G_2-invariant trilinear form. Computational pipeline: 6,560 Zorn matrices $\to$ 9,565 cubic forms $\to$ 47,961 cubic rings, all with discriminant $\equiv 0 \pmod 4$.

9.5. **The magic square.** The Freudenthal-Tits magic square [10] maps pairs of division algebras to Lie algebras via $M(A, B) = \mathrm{Der}(A) \oplus (A_0 \otimes J_0) \oplus \mathrm{Der}(J)$ ($J = J_3(B)$):

	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	$\mathbb{O}$
$\mathbb{R}$	A_1	A_2	C_3	F_4
$\mathbb{C}$	A_2	$A_2 \oplus A_2$	A_5	E_6
$\mathbb{H}$	C_3	A_5	D_6	E_7
$\mathbb{O}$	F_4	E_6	E_7	E_8

The octonionic column produces all five exceptional Lie algebras. $M(\mathbb{O}, \mathbb{O}) = E_8$ connects to the E_8 lattice (Viazovska [27]) and, speculatively, to exceptional Langlands.

10. Machine verification

Every theorem in this paper is formalized in Lean 4 with Mathlib.

Table 3. Key files in the Lean formalization

File	Content	Key Theorems
`Algebra.lean`	Octonion/quaternion integer definitions	—
`OctonionicFermat.lean`	Cubic verification (`native_decide`)	Thm. 3.1
`SymbolicFermatGeneral.lean`	Power reduction (kernel-checked)	Thm. 4.1
`AllOddPrimes.lean`	All odd p (kernel-checked)	Thm. 4.2
`DimensionThreshold.lean`	Dimension boundary	Thm. 5.1
`ZeroDivisorFermat.lean`	Zero-divisor collapse (kernel-checked)	Thm. 7.1
`SedenionZeroDivisor.lean`	Sedenion obstruction analysis	Remark 7.3
`G2OrbitCollapse.lean`	G_2 orbit analysis	Thm. 8.1
`Bridge.lean`	Langlands stratification	Thm. 9.1
`Conjectures.lean`	17 conjectures (documentation)	—

Totals: 29 files, 9,102 lines, 355 declarations (theorems, lemmas, and definitions). The formalization is available at https://github.com/awktavian/fermat (branch `main`, Lean toolchain `v4.16.0`, Mathlib `2026-03-20`).

The formalization uses two proof strategies:

(1) **Kernel-checked symbolic proofs** for the main theorems (Theorems 4.1, 4.2, 7.1). These use only ring axioms and structural induction, with no `native_decide` and no `sorry`. They are valid in any ring satisfying the hypotheses.
(2) **Computational verification** (`native_decide`) for concrete instances: specific triples in $\mathbb{H}_\mathbb{Z}$ and $\mathbb{O}_\mathbb{Z}$, all 21 family verifications, and the Gaussian integer exhaustive search. These verify that the abstract hypotheses are satisfied by the concrete algebras. The sedenion analysis (Section 7) is partially formalized; the obstruction $\alpha^2 = -2$ was identified during formalization.

Zero `sorry`. Zero axioms beyond Lean's foundations (`propext`, `Quot.sound`, `Classical.choice`).

11. Open problems and conjectures

Ten conjectures, selected from 17 in `Conjectures.lean`. All are precise and falsifiable.

Conjecture 11.1 (Artin Confinement — A4)**.** Every solution to $a^p+b^p = c^p$ in $\mathbb{O}_\mathbb{Z}$ (p odd, ≥ 3) lies in an associative subalgebra $\cong \mathbb{H}$.

Evidence. Over 600,000 solutions checked; zero have nonzero associator. The orbit analysis found 61,488 solutions at $\mathrm{N}() \leq 8$, all with $[a, b, c] = 0$. *Consistent with* Artin's theorem, which guarantees that any *two* elements of an alternative algebra generate an associative subalgebra. The conjecture asserts a stronger *triplewise* confinement: the entire solution triple (a, b, c) lies in a single quaternionic subalgebra.

Conjecture 11.2 (Waring Tightness — B1)**.** $g(3, \mathbb{H}_\mathbb{Z}) = 4$.

Evidence. Verified for all 4,785 Lipschitz integers with $\mathrm{N}(q) \leq 30$. Exactly 12 elements (all at norm 26) require 4 cubes.

Conjecture 11.3 (Split G_2 Parametrization — D1)**.** Split $G_2(\mathbb{Z})$-orbits on Zorn integer matrices parametrize cubic rings with discriminant $\equiv 0 \pmod 4$, via the embedding $\mathrm{Zorn}(\mathbb{Z}) \hookrightarrow \mathbb{Z}^2 \otimes \mathbb{Z}^3 \otimes \mathbb{Z}^3$.

Evidence. 6,560 Zorn matrices yield 9,565 cubic forms and 47,961 cubic rings, all with discriminant $\equiv 0 \pmod 4$, all associative.

Conjecture 11.4 (G_2 Fourier Coefficients and Fermat Families — E4)**.** The theta lift of automorphic forms on G_2^c to GSp_6 produces Siegel modular forms whose Fourier coefficients encode the number of octonionic Fermat solution families at each norm level.

Conjecture 11.5 (G_2 Selmer Averaging — E2)**.** $\operatorname{avg}|\mathrm{Sel}_2(G_2, V)| = 3$, by analogy with the Bhargava-Shankar theorem for elliptic curves.

Status. Speculative. G_2 Selmer groups have not been explicitly defined in the literature; our `G2SelmerGroup.lean` is a first attempt.

Conjecture 11.6 (Hasse Principle for $V(\varphi)$ — C2)**.** For the primitive subfamily $V(\varphi)_{\mathrm{prim}}$ (where $a \times b \neq 0$ and a, b, c are pairwise non-proportional), the Hasse principle holds: $V(\varphi)_{\mathrm{prim}}(\mathbb{Q}) \neq \varnothing$ if and only if $V(\varphi)_{\mathrm{prim}}(\mathbb{Q}_v) \neq \varnothing$ for all places v.

Evidence. No local-global failures found; the octonionic cross product is surjective modulo every prime tested ($p \leq 13$), so no Brauer-Manin obstruction at finite primes.

Conjecture 11.7 (Sedenion Fermat — S3)**.** Does the Fermat equation $a^n + b^n = c^n$ have nontrivial solutions in $\mathbb{S}_{\mathbb{Z}}$ for $n \geq 3$? Mechanism A gives odd-exponent solutions (via anti-commuting pairs in quaternionic subalgebras). Mechanism B fails (Section 7). The even-exponent case is open.

Conjecture 11.8 (Waring Linear Growth — B2)**.** $g(p, \mathbb{H}_{\mathbb{Z}}) \sim p + 1$ for odd primes p.

Evidence. Three data points: $g(3) \leq 4$, $g(5) \leq 5$, $g(7) \leq 6$.

Conjecture 11.9 (Zariski Density — C3)**.** $V(\varphi)(\mathbb{Q})$ is Zariski-dense in $V(\varphi)$.

Argument. The fiber structure $\varphi(a, b, c) = \langle c, a \times b \rangle$ gives codimension-1 linear fibers in c for each (a, b) with $a \times b \neq 0$. Dense base and dense fibers imply dense total space.

Conjecture 11.10 (φ-Contraction as G_2 Curvature — F3)**.** The symmetrized contraction identity for the octonionic trilinear form φ (proved algebraically in `TrilinearVariety.lean`) encodes the curvature of G_2-holonomy manifolds in the sense of Bryant [4] and Joyce [16].

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